

Physics-Aware Action-Conditioned World Modeling for Billiards

DiT + Diffusion Forcing on Large Synthetic Multi-Run Datasets

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Abstract

This project studies whether diffusion transformers can preserve physics in an interactive, action-conditioned billiards simulator, and whether diffusion forcing improves long-horizon rollouts. We build a full pipeline: simulation data, VAE latent compression, DiT world-model training, rollout evaluation, and interactive inference tools. The core question is whether multi-step diffusion forcing preserves game logic better than single-step diffusion training. Across multiple runs and model scales, diffusion forcing strongly improves long-horizon stability. Remaining failure modes are mostly identity consistency (color drift and occasional object swallowing), which motivates targeted follow-up losses.

1 Introduction

World models predict future observations conditioned on actions, enabling simulation and interactive generation. Recent diffusion-transformer systems can look realistic, but preserving physical logic over rollout is still hard. We focus on billiards because it gives explicit checks: do balls bounce correctly, collide correctly, and remain consistent identities over time?

Our novelty is to treat *interactive physics consistency* as the main target, not only visual quality. We test single-step DiT versus DiT+diffusion forcing under the same data and evaluation pipeline, and we inspect both metric trends and failure modes.

2 Related Work

Our approach builds on DDPM [5], DDIM [9], latent diffusion [8], DiT [7], and VAE latent modeling [6]. We also draw motivation from world models [3, 4] and action-conditioned diffusion gameplay

work such as Oasis [2, 1].

3 Problem Setup

At each time step t , we observe:

$$x_t \in \mathbb{R}^{72 \times 128 \times 3}, \quad a_t \in \mathbb{R}^3, \quad s_t \in \mathbb{R}^{16 \times 4}, \quad (1)$$

where $a_t = [f_x, f_y, \text{trigger}]$ and s_t stores per-ball simulator state (position and velocity channels).

We train a world model to generate $x_{t+1:t+H}$ conditioned on past context and future actions. In practice we work in latent space $z_t \in \mathbb{R}^{4 \times 9 \times 16}$ for efficiency.

4 Data and Pipeline

4.1 Data generation

We generated three dataset lineages with different goals:

- **v1** (100,000 episodes): baseline coverage.
- **v2** (100,000 episodes): more diverse setups and trajectories.
- **v3** (200,000 episodes): collision-heavy distribution for hard-contact dynamics.

Each episode has 600 frames, for 400,000 episodes total. This staged dataset design was central to improving robustness.

4.2 VAE stage

Frames are compressed by a VAE to 4-channel 9×16 latents (about $48\times$ element compression). We use this latent space for all world-model training and decode for evaluation and visualization.

4.3 Action-conditioned diffusion model

Given context length L , the model receives $z_{t-L+1:t}$, action a_t , and noisy target latent $x_t^{(\tau)}$, and predicts noise $\hat{\epsilon}_\theta$.

Forward diffusion:

$$x_\tau = \sqrt{\bar{\alpha}_\tau} x_0 + \sqrt{1 - \bar{\alpha}_\tau} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \quad (2)$$

Single-step baseline loss:

$$\mathcal{L}_{\text{base}} = \mathbb{E} [\|\hat{\epsilon}_\theta - \epsilon\|_2^2]. \quad (3)$$

4.4 Diffusion forcing

For rollout length K , we train over multiple future steps:

$$\mathcal{L}_k = \|\hat{\epsilon}_k - \epsilon_k\|_2^2, \quad \mathcal{L}_{\text{DF}} = \frac{1}{K} \sum_{k=1}^K \mathcal{L}_k. \quad (4)$$

Predicted clean latent estimate:

$$\hat{x}_{0,k} = \frac{x_{\tau,k} - \sqrt{1 - \bar{\alpha}_\tau} \hat{\epsilon}_k}{\sqrt{\bar{\alpha}_\tau}}. \quad (5)$$

Context for the next step is updated with scheduled teacher forcing:

$$\tilde{z}_{k+1} = m_k z_{k+1}^{\text{gt}} + (1 - m_k) \hat{x}_{0,k}, \quad m_k \sim \text{Bernoulli}(p_{\text{tf}}(n)), \quad (6)$$

where $p_{\text{tf}}(n)$ decays over processed samples n . This reduces train-test mismatch by exposing training to self-generated contexts.

5 Experimental Setup

5.1 Model and training

We trained multiple scales, culminating in a 1.52B-parameter DiT+DF model ($d_{\text{model}} = 2048$, 30 layers, 32 heads) on 2x H100 using distributed data parallel training.

5.2 Evaluation protocol

We use fixed-horizon rollout eval with latent and frame metrics at horizons $h \in \{1, 4, 8, 16, 32\}$:

- latent MSE and latent MAE
- frame PSNR and frame L1

Inference sampling uses DDIM [9] for practical speed/quality tradeoffs.

6 Results

6.1 Baseline vs diffusion forcing

The clearest result is baseline versus DF at context length 8:

Model	MSE@16	MSE@32	PSNR@16	PSNR@32
DiT baseline	54.023	318.312	17.133	14.857
DiT + DF	0.763	0.827	19.447	19.372

Table 1: Long-horizon rollout quality (lower MSE better, higher PSNR better).

DF improves long-horizon stability by a large margin. Qualitatively, trajectories and collisions remain coherent for longer rollouts.

6.2 Scaling and checkpoint progression

Scaling from mid-size to larger models improved short- and mid-horizon quality, though not all long-horizon metrics were monotonic.

6.3 Interactive physics findings

In interactive tests, wall-bounce behavior was generally correct and collision behavior improved substantially after DF plus v3 collision-heavy data. The main remaining issue is identity consistency: some balls still drift in color or occasionally swallow/merge under complex contact. This suggests dynamics are learned better than persistent appearance identity.

Qualitative media is included as supplementary GIFs: VAE reconstruction comparison and latest model rollout preview.

7 Novelty and Main Discovery

Our novelty is the explicit hypothesis test in a controlled physics game:

Can diffusion transformers preserve physical logic better when trained with diffusion forcing rather than single-step diffusion loss?

Our experiments suggest yes, especially at long horizons where error accumulation dominates. Diffusion forcing trains under conditions closer to inference and improves rollout robustness in interactive physics scenarios.

8 Limitations and Future Work

8.1 Current limitations

- Identity consistency (color/object permanence) remains weaker than motion prediction.
- Interactive inference is still slower than true real-time gameplay.
- Distillation path is still under stabilization.

8.2 Planned next steps

- Add color-consistency auxiliary loss on predicted x_0 (ball-focused, simulator-mask-aware).
- Increase collision-focused sampling while keeping uniform sampling mix.
- Complete student distillation for lower-latency interactive inference.
- Evaluate DDIM-step and horizon tradeoffs for playable experience.

9 Conclusion

This project demonstrates that DiT+DF is an effective direction for action-conditioned billiards world modeling. The main takeaway is that diffusion forcing materially improves long-horizon physical stability versus single-step training. We reached credible wall-bounce and collision behavior, while identifying remaining bottlenecks in identity consistency and inference speed. The next phase is targeted identity losses and stabilized distillation for faster interactive rollout.

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